

Exam Project

EXAM FORMAT

The exam for *Introduction to Web Applications* consists of two closely connected parts, both of which are mandatory:

1. Individual design and development of a web application
2. Demonstration and oral discussion of the project.

The evaluation of the projects and the demonstration/oral discussion will concern the material submitted before the exam session. The demonstration and brief oral discussion regarding the project must be carried out according to the schedule that will be made available a few days before the exam session, after registration for the exam session itself.

The discussion will focus on the design choices, such as layout, code structure, database structure, etc., as well as the implementation and functional choices adopted.

DESCRIPTION AND REQUIREMENTS

The project consists of a web application that must satisfy a number of technical, stylistic, and functional requirements, detailed below.

The web application must use the technologies illustrated and tested during the course.

Logistical Requirements

- The project must be carried out individually.
- A submission that is partially or entirely shared with another student enrolled in the course is neither allowed nor acceptable.
- The project must be submitted according to the deadlines indicated on the last page of this document.
- Late submissions are neither allowed nor acceptable.

Technical requirements

The web application must comply with the following technical requirements:

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- Use of HTML5 and CSS3, relying, if necessary, on external frameworks such as Bootstrap, while customizing their style through ad hoc rules.
 - Use of Flask and SQLite, as a relational database, for the back end.
 - Use of Flask-Login for authentication management.

In addition:

- The web application must have a clearly defined target device type, chosen by the students: desktop, mobile, or tablet, optionally supporting responsive design.
- Any forms used in the project must be validated both on the front end and on the back end, as explained during the course.
- The submitted project must be fully testable by the instructor and must work on the latest versions of Chrome, 111+, and Firefox, 110+.
- The source code must be well written and include appropriate comments where necessary.
- All the technologies listed above must be integrated cohesively and consistently within a single web application.

Required:

- Deploy the web application on PythonAnywhere: <https://www.pythonanywhere.com/>

Stylistic Requirements

The web application must comply with the following stylistic requirements:

- Use HTML tags semantically, for example, not everything should be a `<div>`.
- Do not use deprecated HTML tags.
- Do not use inline CSS declarations; CSS rules must be kept separate from the semantic structure of the HTML.

In addition, the web application must be sufficiently usable.

Description: Functional requirements

The goal is to create a web application for managing *Free Walking Tours* in a city chosen by the students. Free Walking Tours are guided tours, usually organized around a city, neighborhood, or specific theme, in which participants do not pay a fixed price in advance but may voluntarily reward the guide at the end of the experience. For simplicity, the application will focus on a single city, chosen by the students, but students are free to define the tours, their themes, the places visited, and the general style of the platform.

The web application must support two types of registered users: **guides** and **participants**.

Guides can create and manage the Free Walking Tours offered on the platform. Before acting as a guide, the user must register and log in. During registration, the guide must provide their first name, last name, email address, and the languages they speak. For simplicity, the available languages are limited to the following five options: Italian, English, Spanish, Portuguese, and German. The email address must be unique and will be used to identify the user on the website.

Participants can browse the available tours and reserve a place in one or more tours, **as long as their schedules do not overlap**. Before making a reservation, participants must register and log in. During registration, each participant must provide their first name, last name, and email address. The email address must be unique and will be used to identify the user on the website.

A participant cannot become a guide, and a guide cannot make reservations as a participant.

Each Free Walking Tour is created and managed by a single guide and must include the following mandatory information:

- Title of the tour.
- Guide who created and manages the tour.
- Weekly schedule, specifying the days of the week and the corresponding start time for each selected day. For simplicity, each tour can have at most one start time per day.
- Meeting point.
- Duration, expressed in minutes.
- Language in which the tour is offered.
- Maximum number of participants allowed **per tour**.
- List of stops included in the tour, corresponding to the names of the places visited during the itinerary (**at least 4**).
- Brief description of the tour.
- A set of 5 promotional photos.

The language of each tour must be chosen from the languages spoken by the guide responsible for that tour.

Students are free to choose the themes of the tours. For example, tours may be historical, artistic, architectural, gastronomic, literary, cinematic, musical, political, alternative, or related to local legends and traditions.

Guides can create new tours and modify the information of their own tours as long as no reservation has been made for that tour. Once at least one participant has reserved a place for any date of the tour, the guide can no longer modify the essential information of the tour, such as weekly schedule, meeting point, duration, language, or maximum number of participants.

All tours are visible to any user, including unregistered users. On the application homepage, users will see a brief version of the available tours. Through appropriate filters, users must be able to explore tours according to:

- Date.
- Duration.
- Language.

By clicking on a tour, users must be able to view the full version of the tour, including all the mandatory information listed above.

After logging in, a participant can reserve a place for a tour. Participants reserve a specific date of a tour, selected from the dates available according to the tour's weekly schedule. When making a reservation, the participant must choose the date on which they want to join the tour.

By default, the reservation is made for the participant themselves. However, the participant may also add up to 3 additional people to the same reservation by providing their names (first name and last name). Therefore, a single reservation may include from 1 to 4 people. The system must prevent reservations that exceed the maximum number of participants allowed for the selected tour date. For example, if a tour has only 2 available places left on a given date, a participant cannot make a reservation for 3 or 4 people.

Reservations can be cancelled by participants only if the cancellation is made at least 24 hours before the start time of the selected tour date. After this deadline, the reservation can no longer be cancelled.

Once a reservation has been completed, it must be visible in the participant's profile page. The profile page must show the tours reserved by the participant, including the date, start time, meeting point, number of people included in the reservation, and the names of any additional participants.

Guides must have a profile page where they can view all the tours they have created. For each tour, guides must be able to see the list of reservations and the total number of expected participants for each scheduled date.

Since guides must pay a commission to the platform for the tours that have actually taken place, the application must also include a post-tour reporting functionality. For each tour date that has already taken place and had at least one reservation, the guide must be able to declare the number of participants who actually attended and upload one photo showing the participants as evidence that the tour was carried out.

Note: upon submission, the web application must contain some sample data allowing the instructors to test its main functionalities. In particular, it must include at least two guides, three participants, several tours with different schedules, languages, and durations, and some

reservations. The usernames and passwords of the sample accounts must be provided to the instructors for verification.

🔥 Students are encouraged to fully express their creativity in choosing the city, the tour themes, the visual identity of the platform, the places included in the itineraries, and the overall experience offered to users. The goal is to create an original and personalized project.

OPTIONAL EXTRA REQUIREMENT FOR PROVA FINALE

Students who need to complete the Prova finale may extend the project by implementing an additional platform administrator profile. For simplicity, the application will include only one platform administrator account.

The platform administrator does not create tours and does not make reservations. Instead, he has access to a dedicated administration page where he can view general information about the platform.

In particular, the administrator must be able to:

- View the list of all registered guides.
- View the main information associated with each guide, such as first name, last name, email address, and spoken languages.
- View the tours created by each guide **with all its detailed information**.
- View basic statistics about the platform, such as:
 - total number of guides;
 - total number of participants;
 - total number of tours;
 - total number of reservations;
 - total number of reservations per language.

This additional functionality is optional and **is required only for students completing the Prova finale**. It should be integrated into the same web application in a coherent way, without replacing or simplifying the mandatory requirements described above.

PROJECT SUBMISSION INSTRUCTIONS

The project must be submitted through the “**Consegna Elaborati**” tool available on the course page on the Teaching Portal, by **23:59 on 28 June 2026**.

The submission must contain a single archive file, **.zip**, including:

- Source code of the developed web application, including:
 - any dependencies, in a file such as **requirements.txt**;
 - images;
 - etc.;

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- the SQLite file containing the database.
 - A text document, `.md` or `.txt`, containing:
 - the user credentials;
 - any instructions needed to test the web application;
 - the url where the deployed web application is available.